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Tamirlan Zharmagambetov

English - Fluent

Russian - Native

Kazakh - Native

B.S. Virginia Polytechnic Institute and State University (Virginia Tech) 2022 - 2026 (GPA: 3.4)

Degree: Computer Engineering

Major: Controls, Robotics, and Autonomy | Minor: Machine Learning and Artificial Intelligence

RELEVANT EXPERIENCE:

Software Engineering Lead | VT CRO - AutoNav Robotics | August 2025 – Present

- Developing a ROS2 autonomous navigation stack for an IGVC-style rover on Jetson Orin Nano.
- Implemented perception for line and obstacle detection using ZED2i, LiDAR, GPS, and IMU fusion.
- Integrated and debugged the full ROS stack, including odometry, motor control, encoders, perception, path-planning, simulation, and automated tests.
- Containerized the system with Docker and improved reliability by stabilizing sensor interfaces.

Computer Engineering Intern | Inertial Labs, a VIAVI Solutions Company | June 2025 – Aug 2025

- Developed C++ firmware on MX8 for LiDAR, camera, IMU, and INS integration.
- Led full-stack static test tool project combining embedded firmware and software system.
- Built a security layer for Rust backend with PostgreSQL API and SSL encryption.
- Developed Kotlin frontend for license management with customer-assignment features.
- Contributed to Flutter app for sensor calibration and firmware updates, adding security layers.

Software Engineering Intern | Basis | May 2024 – Sep 2024

- AI Fine-Tuning & Prompt Optimization: Used both real and synthetic data and Automatic Prompt Optimization with "Gradient Descent" to improve the accuracy and efficiency of LLMs.
- Software Development: Collaborated on software projects in Python, built embedded applications using Retool software, and navigated the SQL databases using complex queries.
- Collaborated with AI experts on building 'Prompt Olympics' to enhance Python script development for creating testing environments and training models in an AI-driven workplace.

Assistant to the Chief Financial Officer | MOST Ventures | Jun 2023 – Aug 2023

- Assisted CFO in evaluating Pre-seed/Seed IT startups across 6 countries.
- Conducted thorough reviews and analyses of startup viability and growth prospects.
- Participated in startup pitching meetings, gaining insights into venture capital dynamics.
- Developed a comprehensive understanding of venture capital operations and startup investing.

RELEVANT PROJECTS:

“Computer Vision Based Real-Time Poker Advisor” | August 2024 – December 2024

- Built a real-time poker table analyzer using Python, OpenCV, and Tesseract OCR to extract the complete game state from live PokerStars tables via resolution-independent ROI screen capture.
- Optimized OCR with preprocessing, character whitelisting, and frame-skipping.
- Implemented template-matching for card recognition using normalized cross-correlation.
- Structured data into a unified table-state model for hand evaluation and GTO decision-making.

“Home Audio System” | August 2024 – December 2024

- Designed a three-band graphic equalizer using fourth-order Butterworth filters (100 Hz–8 kHz).
- Developed a Class-D amplifier with PWM modulation, diode-based deadtime control, and RLC filtering.
- Programmed an Arduino for 80 kHz PWM carrier wave and OLED real-time spectrogram display.
- Conducted circuit simulations and testing using LTSpice and WaveForms
- Integrated and optimized subsystems including the equalizer, amplifier, and microcontroller display.

SKILLS:

C++
Python

PyTorch
ROS2

CV + OCR
Docker

Linux Yocto (MX8)
GNSS/IMU Fusion